PROOF OF CONCEPT REPORT

# Electronic Medicine Approval System with eCTD Technical Validation

تقرير إثبات المفهوم: نظام إلكتروني للموافقة على الأدوية مع التحقق الفني بصيغة eCTD

| | |
|---|---|
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# 1 Executive summary

This report presents a working proof of concept (POC) for an electronic medicine approval system, named PPR. Pharmaceutical companies submit marketing authorisation dossiers in the international eCTD format. The system checks every package automatically, as the FDA and EMA do, before any scientific review starts.

The POC covers the complete process: submission, technical validation, scientific review, requests for information, the Head of Department's decision and a bilingual approval certificate. It operates in English and Arabic and is ready to deploy on PythonAnywhere with a MySQL database.

| | | | |
|---|---|---|---|
| **3** user roles | **10** application states | **20** validation rules in 3 layers | **2** languages (English, Arabic) |
| **44** automated tests passed | | | |

# 2 Background and objectives

## 2.1 Background

The FDA and the EMA receive drug applications as eCTD (electronic Common Technical Document) packages. Each submission is a folder called a *sequence* (0000, 0001, …) containing an XML backbone, `index.xml`, that lists every document in the five CTD modules together with its checksum.

Before the scientific content is assessed, each package must pass eCTD validation: an automated, rules-based technical check that the files are formatted correctly, are not secured, and are structured as required. The check has three layers: XML schema validation, business rule checks and file compliance inspection.

## 2.2 Objectives

- Receive medicine approval applications electronically in eCTD format.
- Apply the three validation layers automatically and give the applicant a clear report.
- Support the scientific review, requests for information and the final decision in one system.
- Keep a complete audit trail of every decision and status change.
- Provide an Arabic and English interface.

# 3 Scope of the proof of concept

## 3.1 In scope

- Company, reviewer and Head of Department roles
- Application creation and eCTD zip upload
- Three-layer technical validation with reports
- Dossier viewer organised by CTD heading
- Module reviews, information requests and response sequences
- Decision, approval number and bilingual certificate
- Audit trail, in-app and email notifications

## 3.2 Out of scope

- Fees and online payment
- Company self-registration and verification
- Integration with other government systems
- Official regional Module 1 specification
- Single sign-on and two-factor login

# 4 Business workflow

Every status change passes through a single state machine. It checks that the user's role is allowed to make the change, records it in the audit trail and notifies the people involved. Figure 1 shows the states and who triggers each transition.

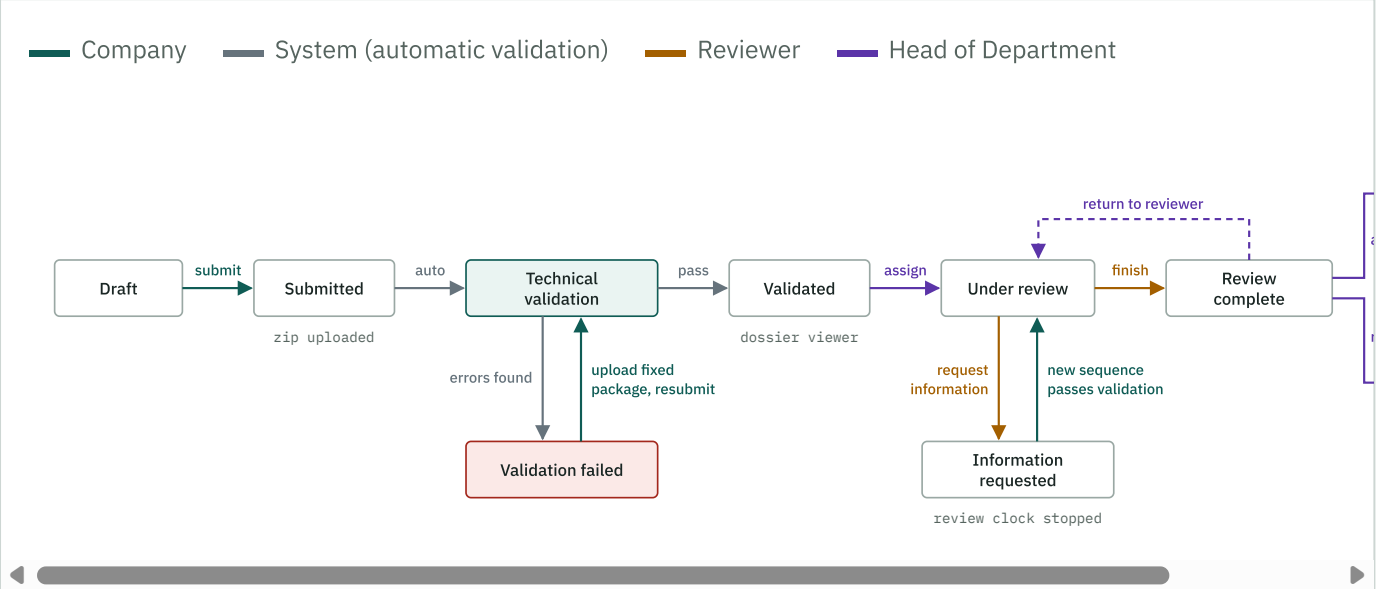



**Figure 1:** Application states and transitions. A package cannot reach a reviewer until it passes validation; answers to reviewer questions are new sequences and are validated in the same way.

| Role | Responsibilities in the system |
|---|---|
| **Company applicant** | Creates applications, uploads eCTD sequences, submits, reads validation reports, answers information requests and prints the certificate. |
| **Reviewer** | Works only on assigned applications. Browses the dossier by CTD heading, records a review per module, requests information and sends a recommendation. |
| **Head of Department** | Monitors workload and status, assigns reviewers, approves or rejects, or returns the file to the reviewer. |
| **System administrator** | Manages user accounts, companies and roles. |

**Table 1:** User roles.

# 5 eCTD technical validation

Validation runs when the company submits. The package is unpacked safely, the XML is read once, and 20 rules run in three layers in a fixed order. Any error fails the sequence; warnings are reported but do not block it.

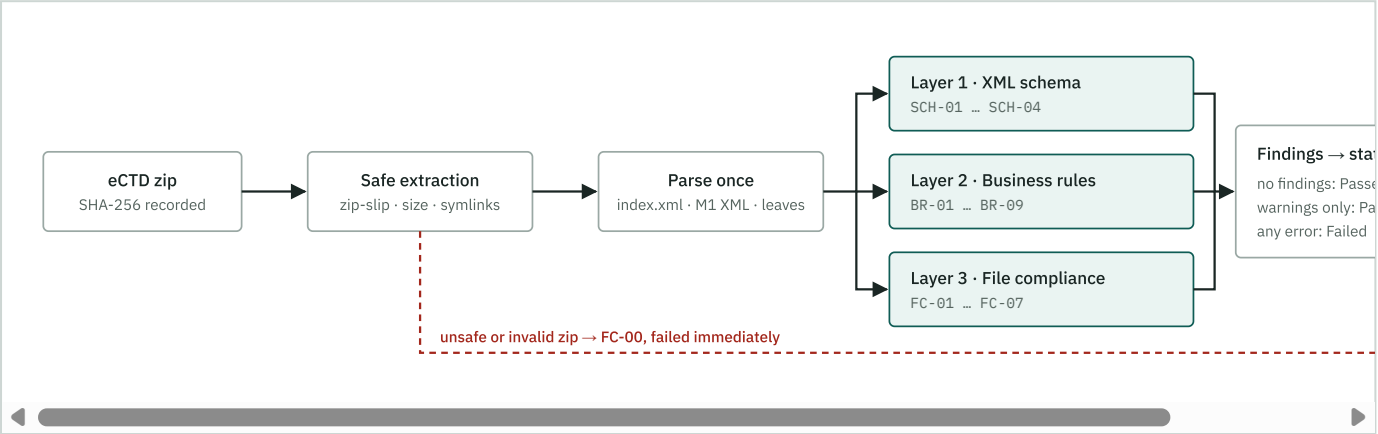



**Figure 2:** Validation pipeline. Failed packages are not kept unpacked; passed sequences become available in the dossier viewer.

| Layer | Rule | Requirement | Severity |
|---|---|---|---|
| **1. XML schema validation** | SCH-01 | The sequence contains `index.xml` | Error |
| | SCH-02 | `index.xml` is well-formed XML | Error |
| | SCH-03 | `index.xml` is valid against the ICH eCTD 3.2 DTD | Error |
| | SCH-04 | Regional Module 1 XML exists and is well-formed | Error |
| **2. Business rule checks** | BR-01 | One sequence folder with the expected 4-digit number | Error |
| | BR-02 | `index-md5.txt` matches the MD5 of `index.xml` | Error |
| | BR-03 | Every document reference points to a file in the sequence | Error |
| | BR-04 | Every document's MD5 checksum matches the declared value | Error |
| | BR-05 | No files that the XML does not reference | Warning |
| | BR-06 | Modules required for the application type contain documents | Error |
| | BR-07 | Replace, append and delete operations refer to earlier accepted sequences | Error |
| | BR-08 | ICH naming: lowercase letters, digits and hyphens; names up to 64 and paths up to 180 characters | Error |
| | BR-09 | `util/dtd` contains the eCTD DTD | Error |
| **3. File compliance inspection** | FC-00 | The package is a valid, safe ZIP archive | Error |
| | FC-01 | Accepted file formats only (.docx in Module 1 only) | Error |
| | FC-02 | PDF files are not encrypted or password-protected | Error |
| | FC-03 | PDF version is 1.4 to 1.7 | Error |
| | FC-04 | PDF files contain no JavaScript or launch actions | Error |
| | FC-05 | Fonts in PDF files are embedded | Warning |
| | FC-06 | No file exceeds the size limit (400 MB) | Error |

| Layer | Rule | Requirement | Severity |
|---|---|---|---|
| | FC-07 | No empty files | Error |

**Table 2:** Validation rules by layer.

# 6 System screens

The following screens were captured from the running prototype using demonstration data. Each image opens at full size.

## 6.1 Company applicant

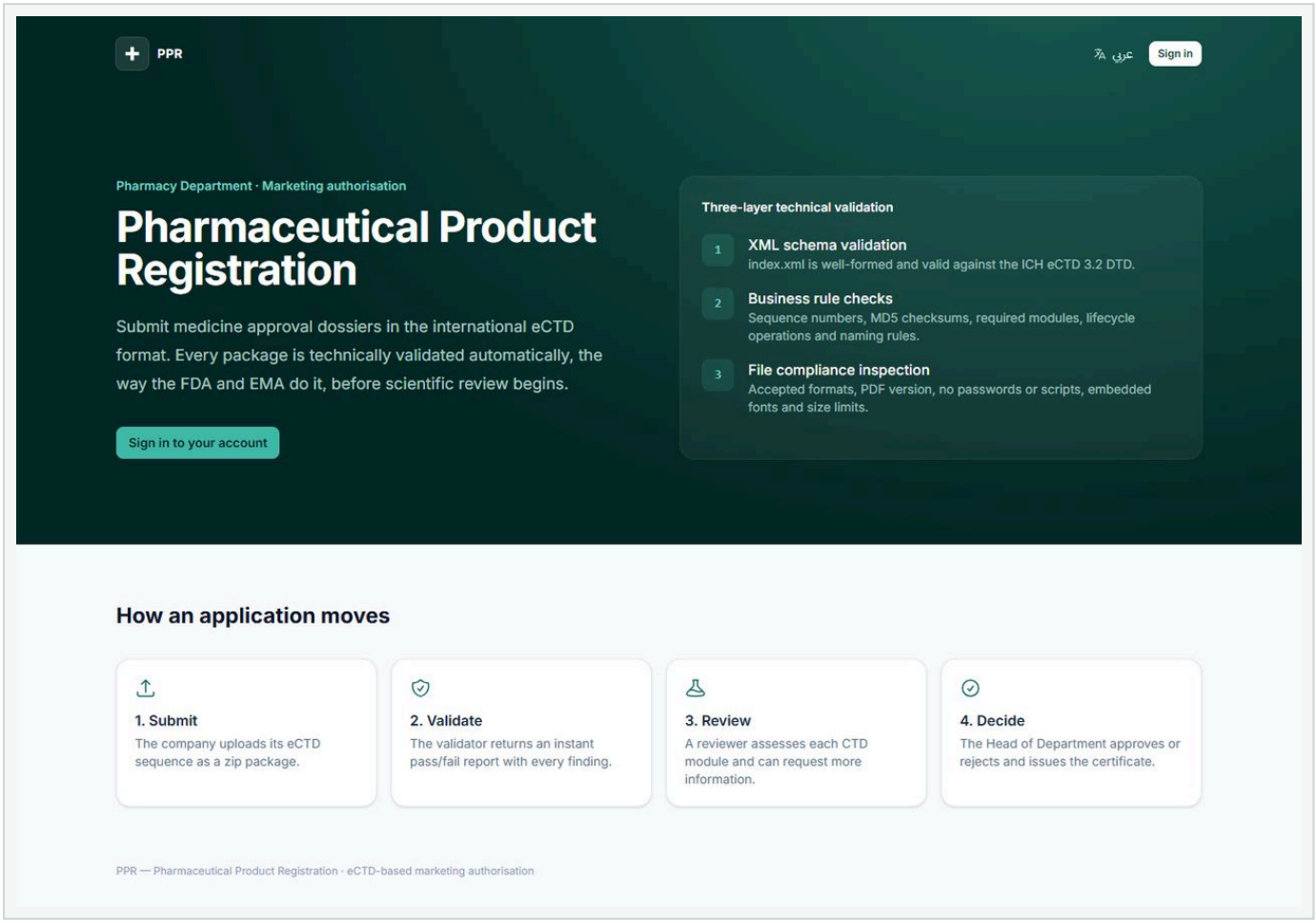



**Figure 3:** Public landing page describing the process and the three validation layers.

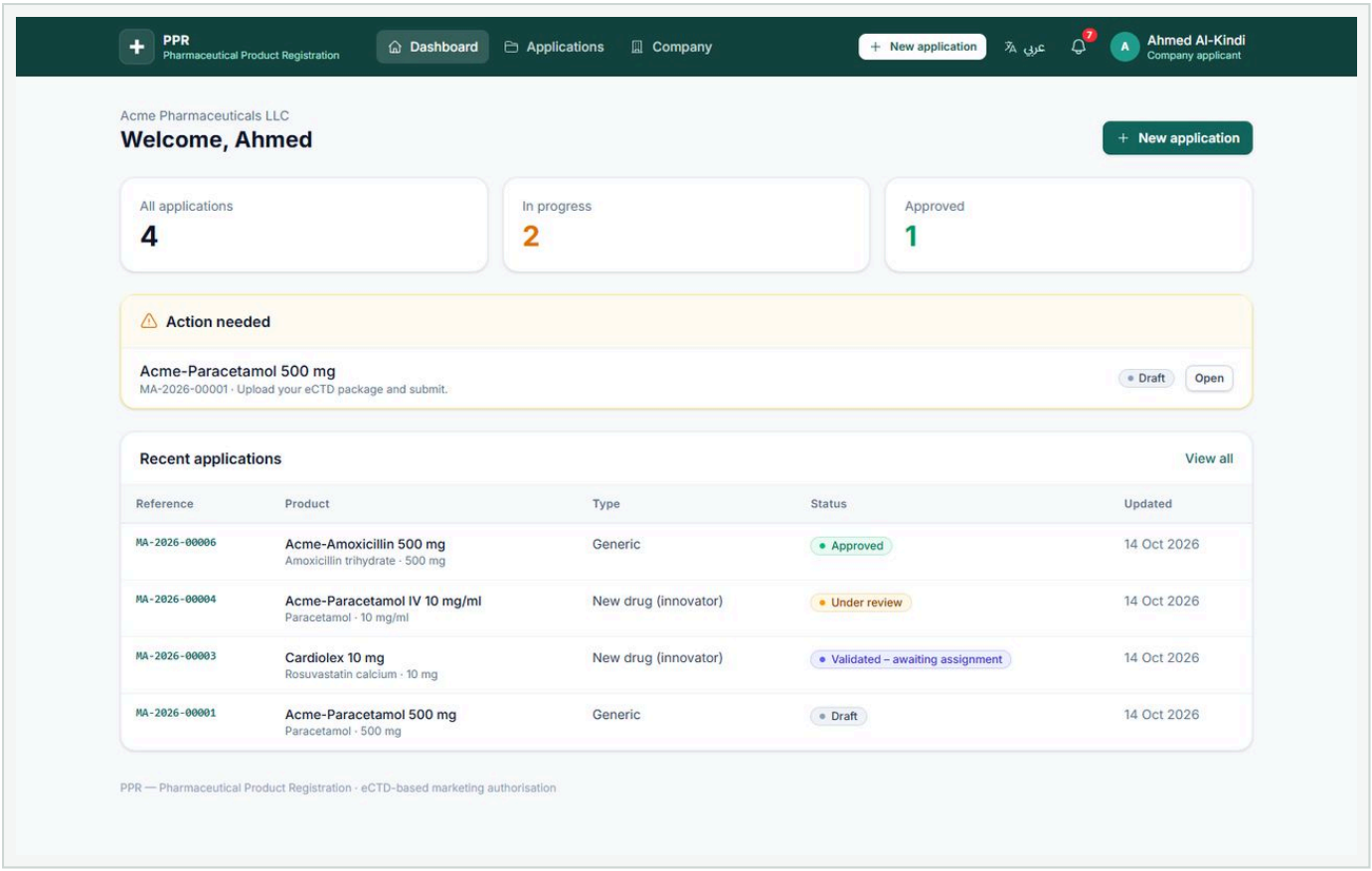



**Figure 4:** Company dashboard with totals, items needing action and recent applications.

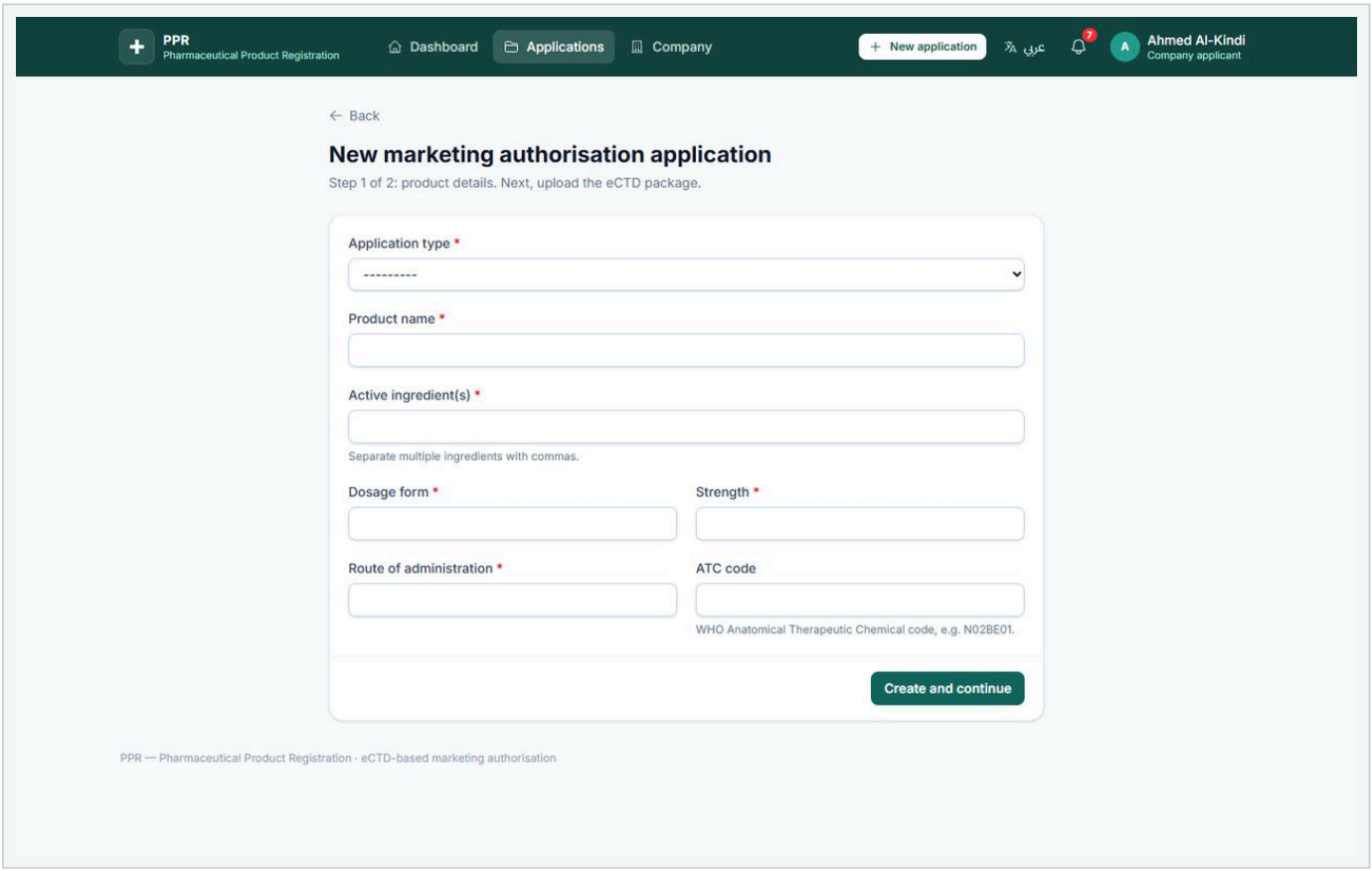



**Figure 5:** New application form for product details.

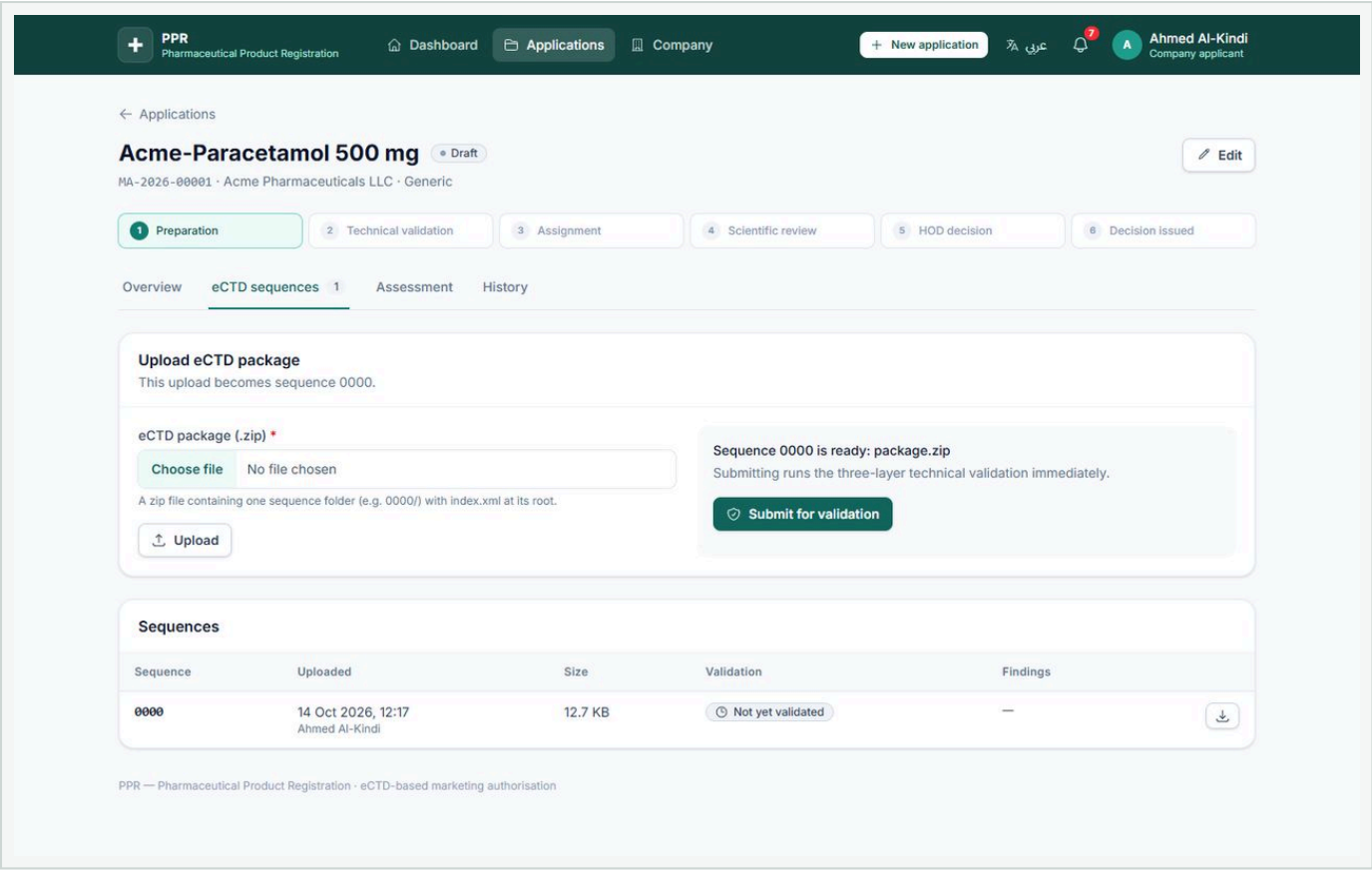



**Figure 6:** eCTD package upload. Submission starts validation immediately.

## 6.2 Technical validation

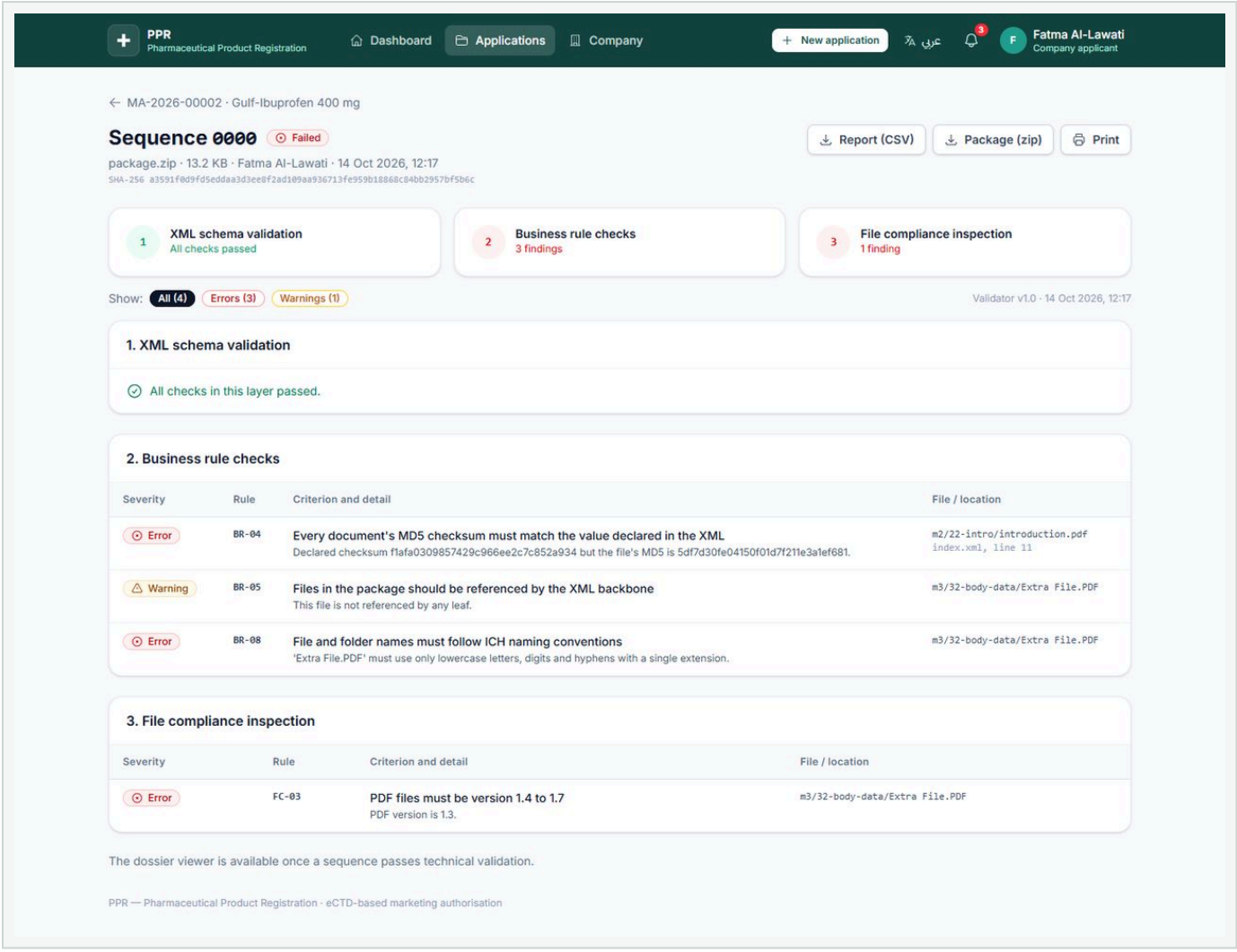



**Figure 7:** Failed validation report listing each finding with its rule, file and line, exportable as CSV.

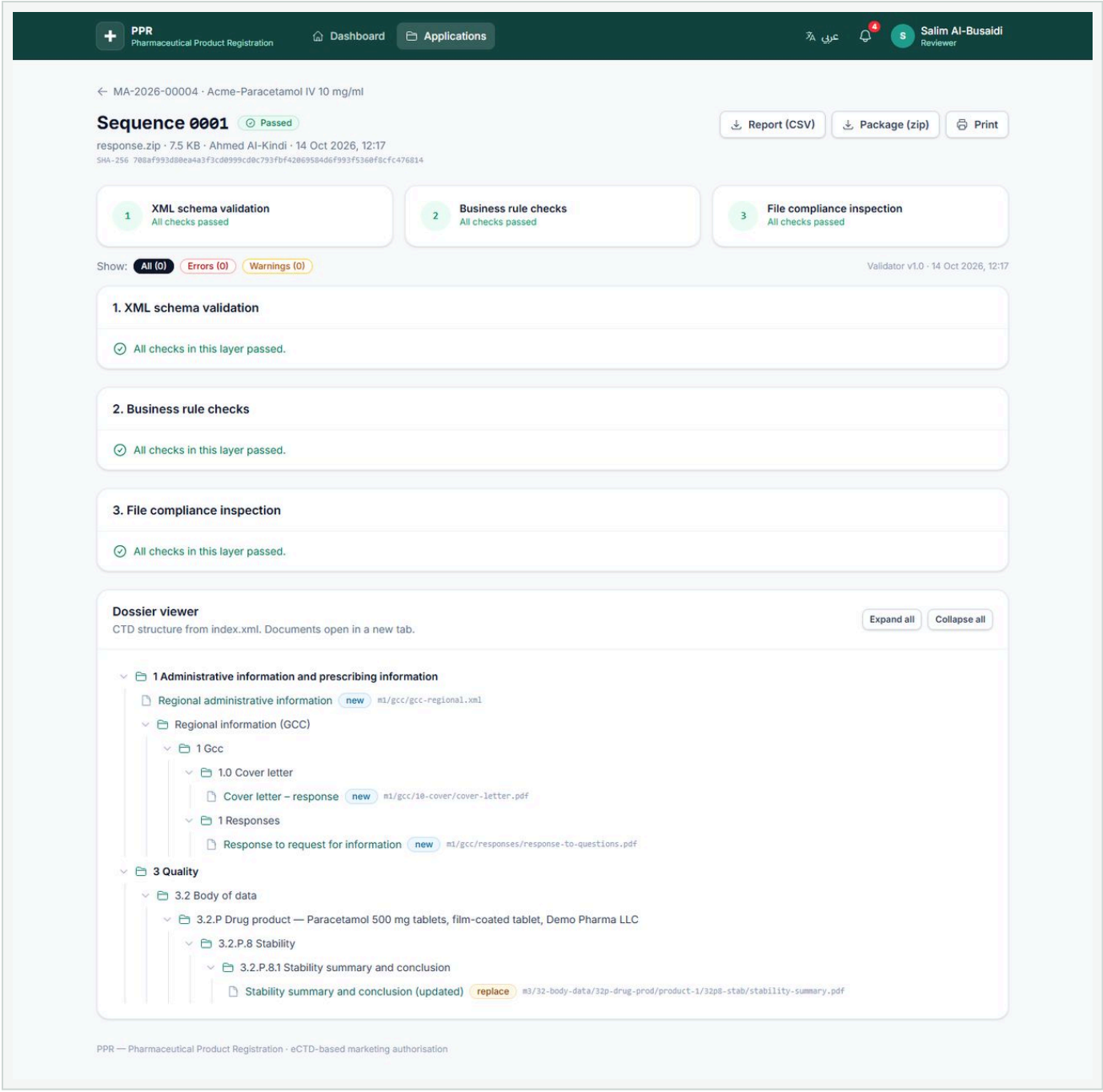



**Figure 8:** Passed report and dossier viewer. Sequence 0001 replaces the stability summary submitted in 0000.

## 6.3 Review and decision

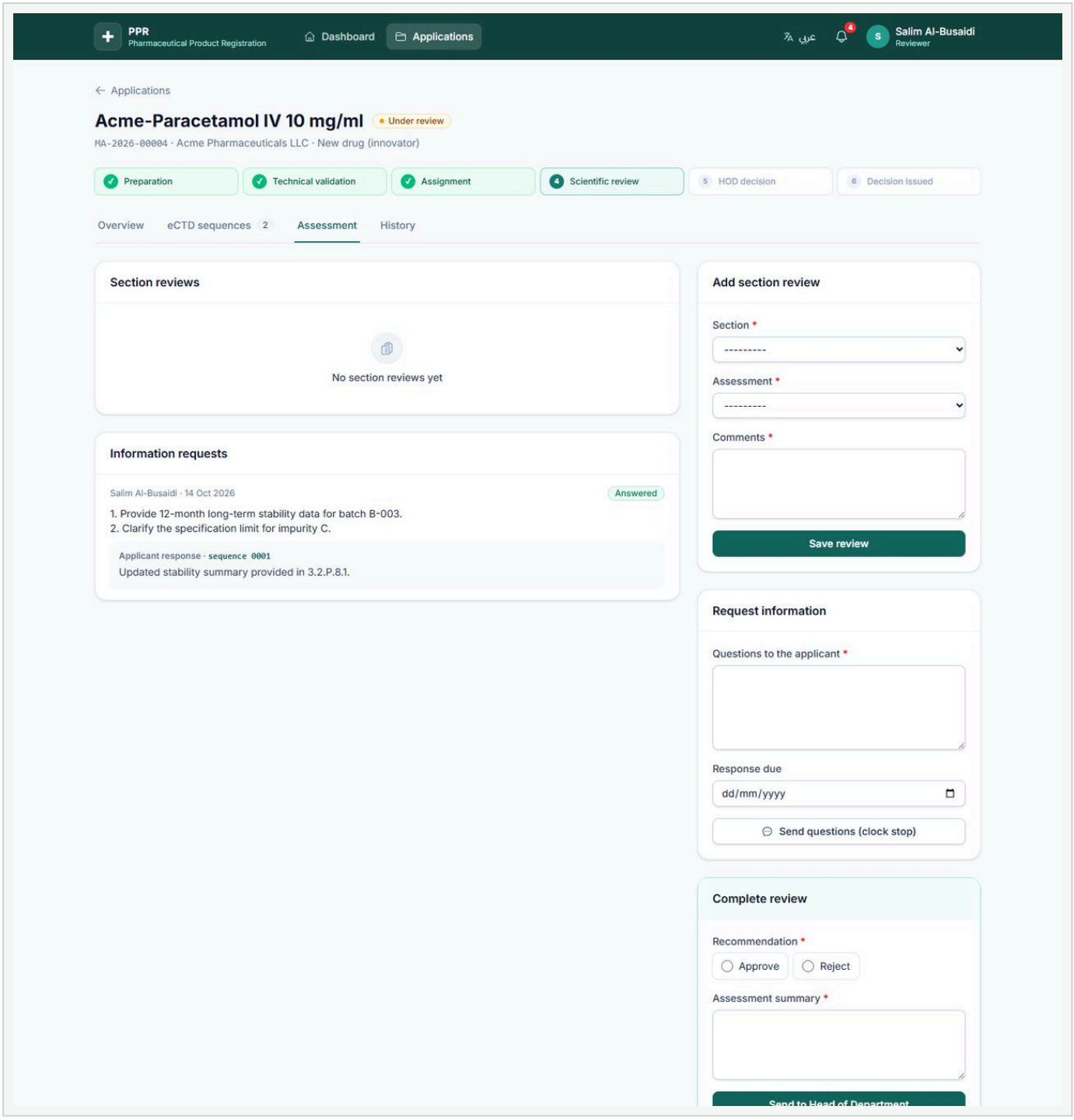



**Figure 9:** Reviewer assessment: module reviews, information request with the company's response, and the recommendation form.

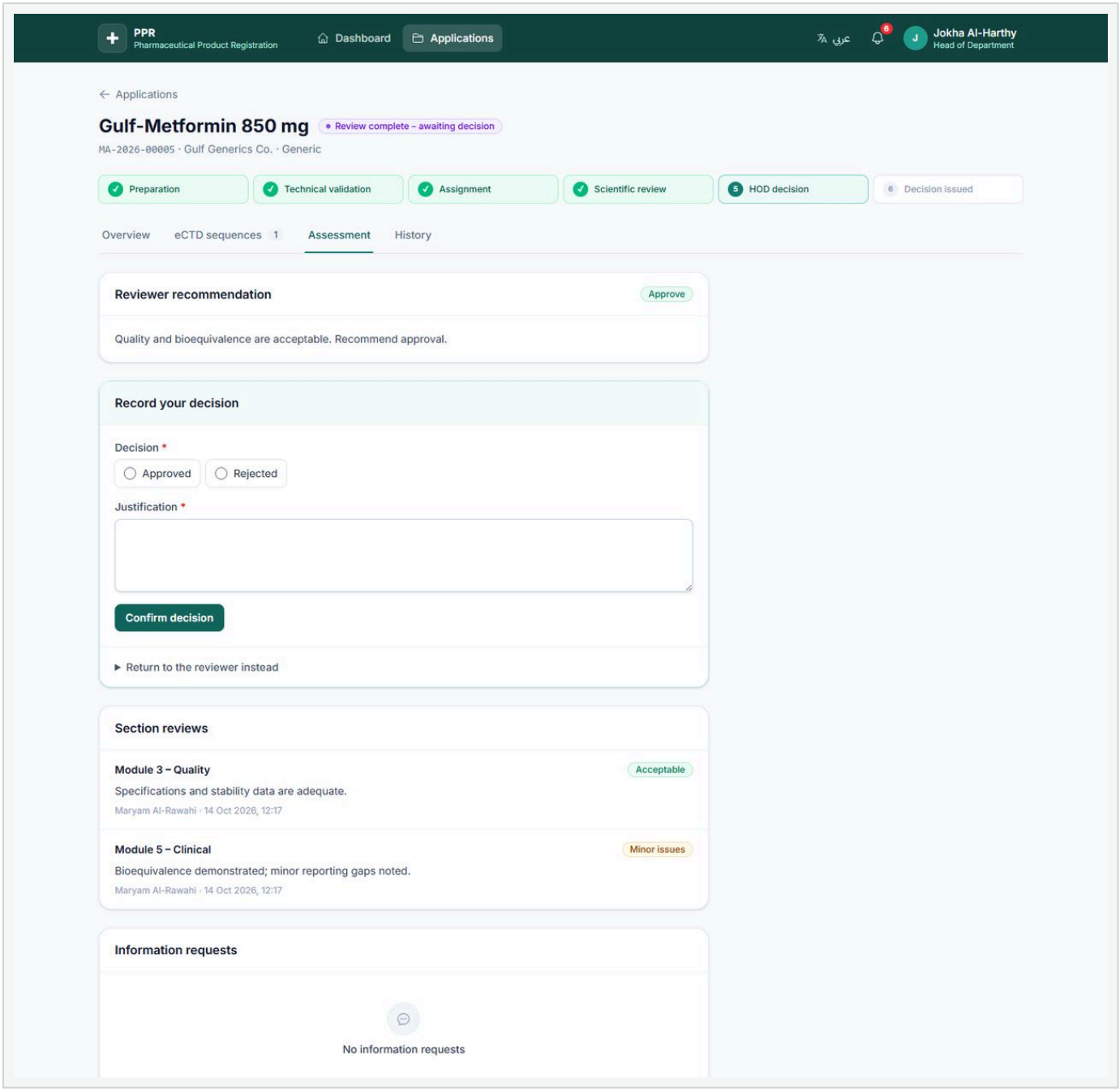



**Figure 10:** Head of Department decision based on the reviewer's recommendation.

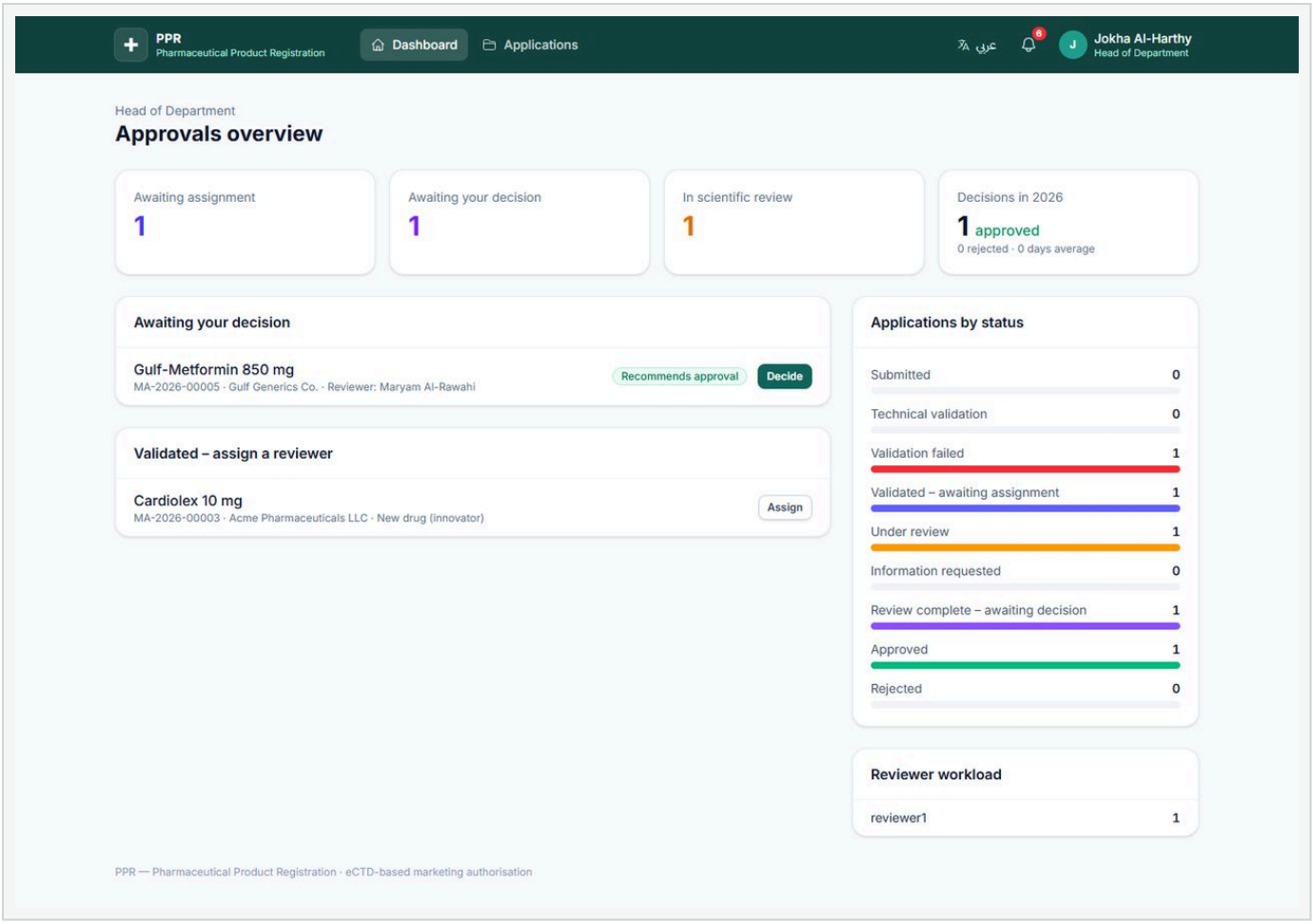



**Figure 11:** Head of Department dashboard: pending decisions and assignments, status summary and reviewer workload.

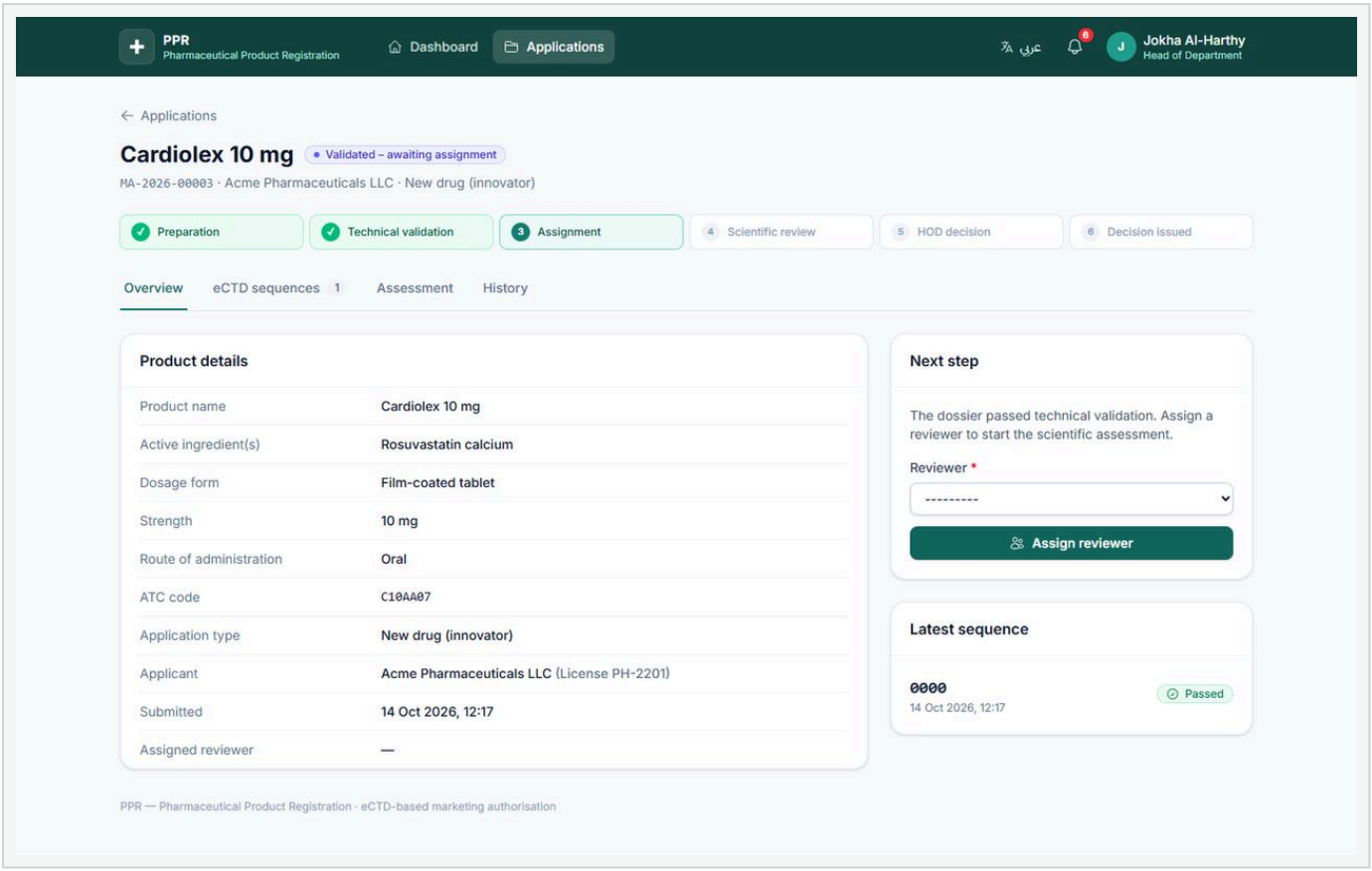



**Figure 12:** Application overview with progress indicator and reviewer assignment.

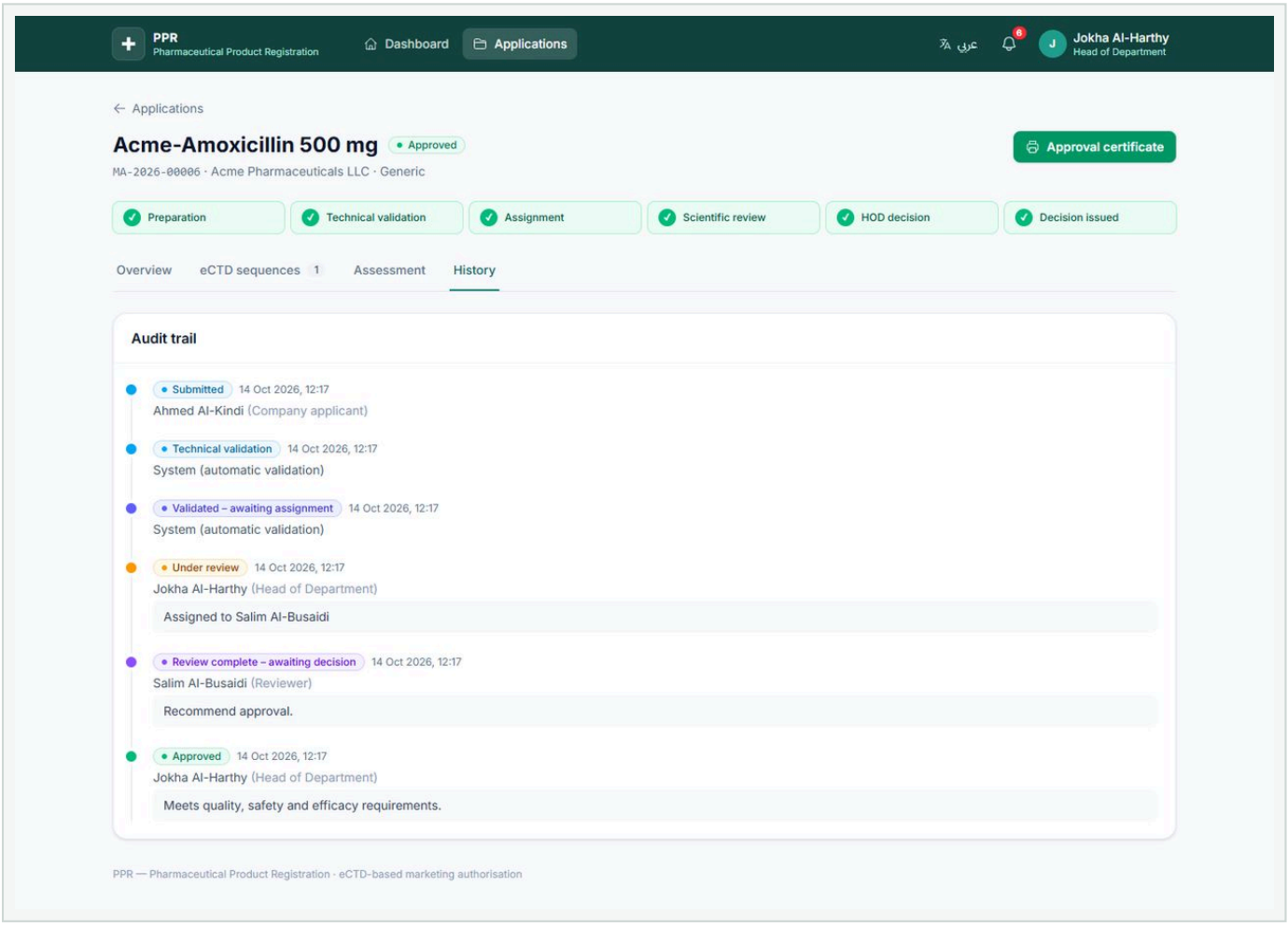



**Figure 13:** Audit trail of every status change, with user, time and note.

## 6.4 Arabic interface and mobile layout

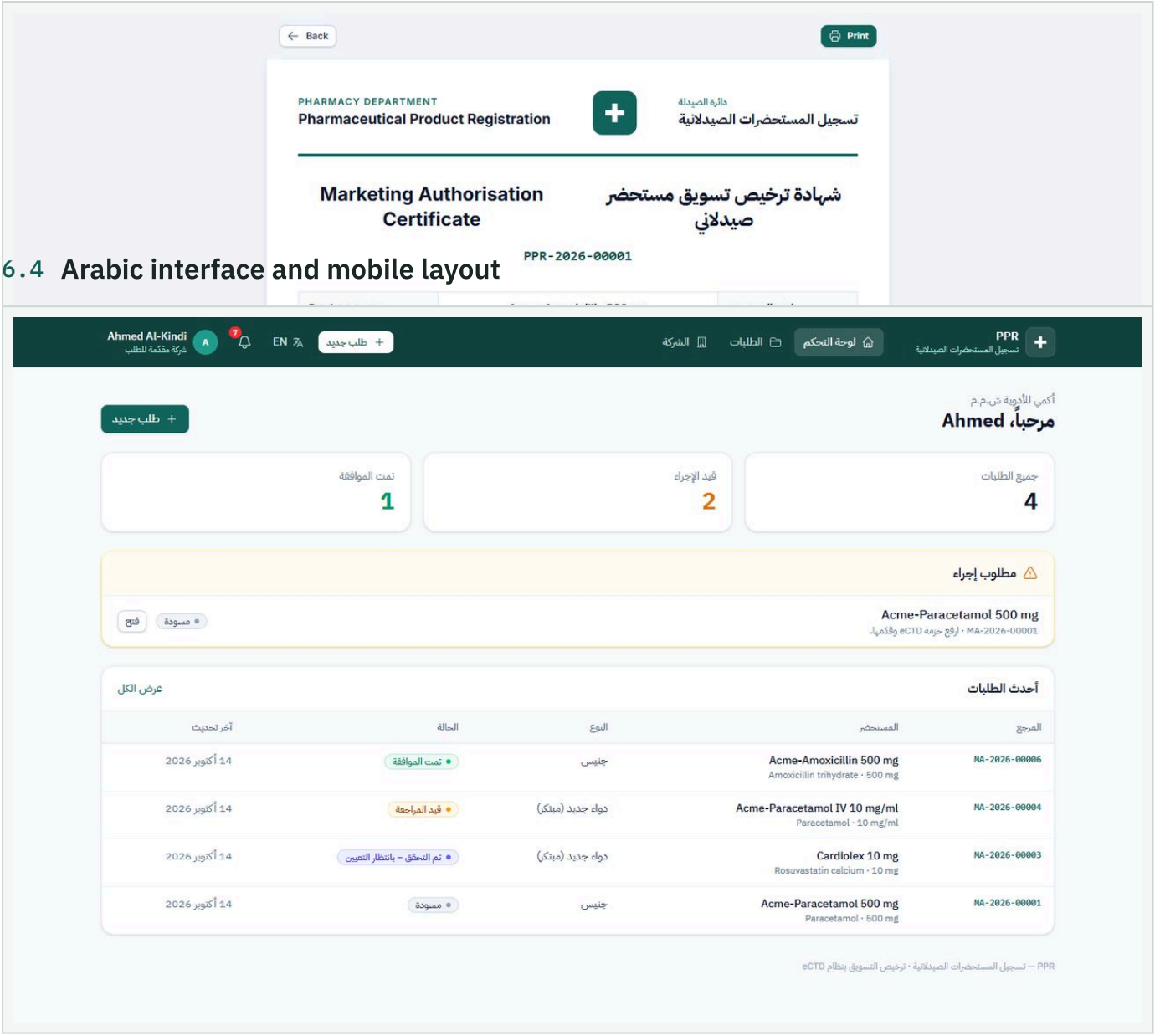



**Figure 15:** Company dashboard in Arabic with a right-to-left layout.

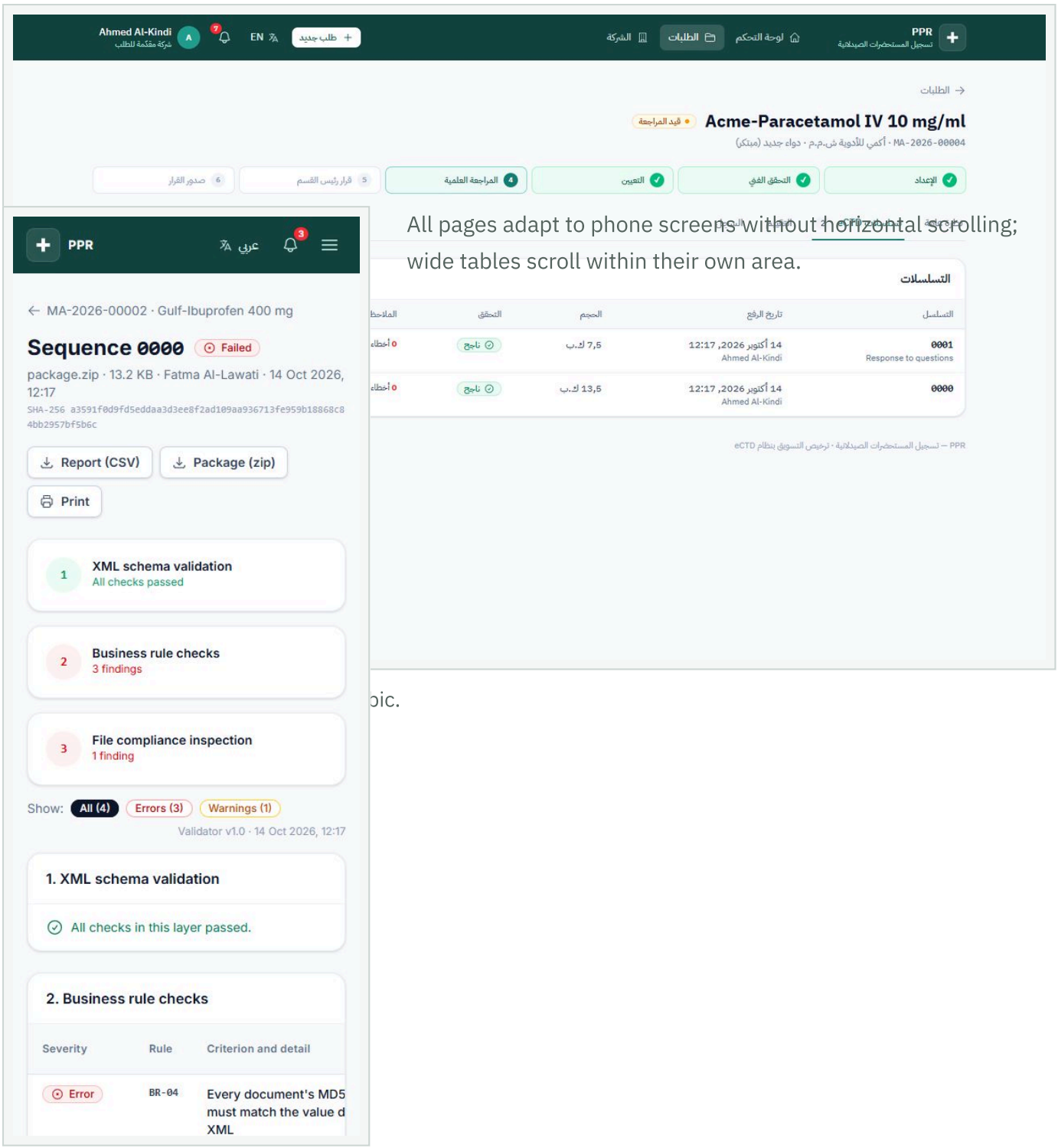



**Figure 17:** Validation report on a phone.

# 7 Technical architecture

PPR is a single Django web application. Validation runs within the submission request, which suits PythonAnywhere's standard plans. Uploaded dossiers are never published under a public address; every download passes through an access check.

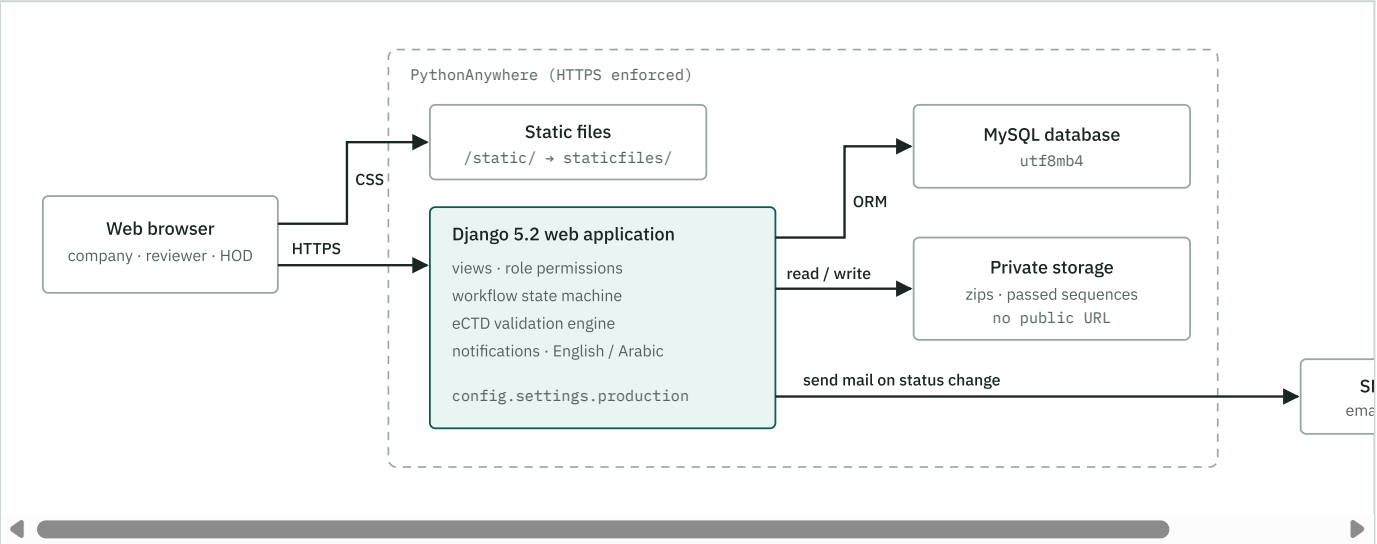



**Figure 18:** Deployment architecture on PythonAnywhere.

| Component | Location in source code | Responsibility |
|---|---|---|
| Workflow | `apps/applications/workflow.py` | Allowed transitions per role, audit trail, notifications |
| Business services | `apps/applications/services.py` | Submit, respond, assign, review and decide |
| Validation engine | `apps/validation/` | Safe extraction, parsing, 20 rules, findings, dossier tree |
| Accounts | `apps/accounts/` | Users with role and company; access control |
| Notifications | `apps/notifications/` | In-app notifications and email in the user's language |
| User interface | `templates/, theme/` | Tailwind templates supporting left-to-right and right-to-left |

**Table 3:** Main software components.

# 8 Technology stack

| Area | Technology | Version | Reason for selection |
|---|---|---|---|
| Programming language | Python | `3.12` | Supported by PythonAnywhere; mature XML and PDF libraries |
| Web framework | Django | `5.2 LTS` | Long-term support until 2028; built-in admin, authentication, translation and security |
| Styling | Tailwind CSS (django-tailwind) | `4.3 / 4.6` | Standalone build without Node.js |
| Interactivity | HTMX, Alpine.js | `2.0.4 / 3.14.8` | Lightweight interactions without a separate front-end application |
| XML processing | lxml | `6.1` | DTD validation with a parser hardened against external entities |
| PDF inspection | pypdf | `6.20` | Checks encryption, version, scripts and fonts |
| Database | SQLite (development), MySQL (production) | `mysqlclient 2.2` | No setup locally; MySQL is provided by PythonAnywhere |
| Configuration | django-environ | `0.14` | Secrets and database settings kept outside the code |
| Localisation | Django i18n with Babel and polib tooling | `EN / AR` | 397 interface strings translated into Arabic |
| Typefaces | Inter, IBM Plex Sans Arabic | `Google Fonts` | Consistent Latin and Arabic text |
| Testing | pytest, pytest-django | `9.1 / 4.14` | Automated tests on an in-memory database |
| Hosting | PythonAnywhere | `WSGI` | Managed Python hosting with HTTPS and MySQL |

**Table 4:** Technology stack of the proof of concept.

### 8.1 Data model

- **Company**: registration and licence numbers, English and Arabic names
- **User**: role and company
- **Application**: product details, type, status, assigned reviewer
- **Sequence**: one eCTD submission with SHA-256 and validation status
- **Validation run and finding**: the result of every rule
- **Review, information request, decision**: assessment and outcome
- **Status history, notification**: audit trail and alerts

### 8.2 Operational tools

- `seed_demo`: demonstration users and applications at each stage
- `build_sample_packages`: valid and invalid eCTD test packages
- `validate_sequence`: re-runs validation, for example after a DTD update
- `scripts/i18n.py`: extracts and compiles translations
- `docs/deploy_pythonanywhere.md`: deployment procedure

## 9 Security and quality assurance

### 9.1 Security controls

- Role checks on every page: companies see only their own applications, reviewers only their assignments.
- Status changes only through the state machine, each recorded in the audit trail.
- Package extraction blocks path traversal, symbolic links, encrypted archives and oversized archives.
- XML read with external entities, DTD loading and network access disabled.
- Dossier documents served only after an access check, with protective security headers.
- Production settings enforce HTTPS, secure cookies, CSRF and clickjacking protection; Django's deployment check passes.

### 9.2 Testing

| Test area | Result |
|---|---|
| Valid packages pass all rules | Passed |
| One invalid package per rule | Passed |
| Full lifecycle, resubmission and response paths | Passed |
| Permissions per role | Passed |
| All pages per role, English and Arabic | Passed |
| Protected downloads and invalid uploads | Passed |
| **Total: 44 automated tests** | **44 passed** |

**Table 5:** Automated test results.

# 10 Limitations and recommendations

**Required before accepting real submissions:** replace the reference DTD included in the POC with the official `ich-ectd-3-2.dtd` from the ICH eCTD v3.2.2 specification, and confirm the regional Module 1 structure (currently `gcc`) against the specification adopted by the department.

| No. | Recommendation | Phase |
|---|---|---|
| R1 | Adopt the official ICH DTD and regional Module 1 specification | Pilot |
| R2 | Run validation in the background for large dossiers | Pilot |
| R3 | Company self-registration with document verification | Pilot |
| R4 | Single sign-on or two-factor login for staff | Pilot |
| R5 | Backup, retention and storage policy for dossiers | Pilot |
| R6 | Combined lifecycle view of all sequences in a dossier | Phase 2 |
| R7 | Review timelines with target days and reports | Phase 2 |
| R8 | QR code on certificates for public verification | Phase 2 |
| R9 | Fees and payments; variations and renewals linked to approvals | Phase 2 |
| R10 | Accessibility review and load testing | Phase 2 |

**Table 6:** Recommendations for the next phases.

## 11 Approval

**PREPARED BY**

Name

Signature

Date

**REVIEWED BY**

Name

Signature

Date

**APPROVED BY · HEAD OF DEPARTMENT**

Name

Signature

Date

# A Glossary

| Term | Meaning |
|---|---|
| **CTD** | Common Technical Document: the ICH structure of five modules (administrative, summaries, quality, non-clinical, clinical). |
| **eCTD** | The electronic form of the CTD, with an XML backbone listing every document. |
| **Sequence** | One eCTD submission, numbered 0000 for the initial dossier and 0001 onwards for later submissions. |
| **DTD** | Document Type Definition: the rules an eCTD `index.xml` must follow. |
| **MD5 checksum** | A fingerprint of a file used to confirm it has not changed. |
| **Lifecycle operation** | How a document relates to earlier sequences: new, replace, append or delete. |
| **ICH** | International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use. |
| **HOD** | Head of Department. |